

4. Bud rate

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1 Environment preparation

Run scripts with the needed packages, functions, and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
source("0c_functions.R")
```

2 Data formatting

Create a new data frame adapted for generalized linear mixed-effects models (GLMM).

```

buds_df <-

  #' Load the raw data frame.
  df %>%

  #' Reshape the weekly bud counts from wide to long format.
  pivot_longer(cols = starts_with("buds_"),
               names_to = c(".value", "week"),
               names_sep = "_",) %>%

  #' Convert week and bud count to integers.
  mutate(week = as.integer(week),
         buds = as.integer(buds),
  #' Create a centered variable for the week.
         week_centered = week - mean(week, na.rm = TRUE)) %>%

  #' Remove weeks with missing bud count.
  filter(!is.na(buds)) %>%

  #' Keep only the variables needed for the analysis.
  select(sleep_condition, lineage, tum_state, week, week_centered,
         buds, id_unique, parent_unique, replicate_unique)

```

Set reference levels for the categorical variables.

```

buds_df$lineage <- relevel(factor(buds_df$lineage), ref = "HO_MT")
buds_df$sleep_condition <- relevel(factor(buds_df$sleep_condition), ref = "Control")
buds_df$tum_state <- relevel(factor(buds_df$tum_state), ref = "Healthy")

```

3 Model selection

3.1 Random effects selection

With the full fixed-effect structure held constant, compare different random-effect structures to find the optimal one.

```

buds_pois_R_00 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition, REML = TRUE,
family = poisson, data = buds_df)
buds_pois_R_01 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_02 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition +
(week_centered | id_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_03 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_04 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_05 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | parent_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_06 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | replicate_unique), REML = TRUE, family = poisson, data = buds_df)
buds_pois_R_07 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
id_unique) + (1 | parent_unique) + (1 | replicate_unique), REML = TRUE, family = poisson,
data = buds_df)

```

```
AIC(buds_pois_R_00, buds_pois_R_01, buds_pois_R_02, buds_pois_R_03, buds_pois_R_04,
    buds_pois_R_05, buds_pois_R_06, buds_pois_R_07) %>% arrange(AIC)
```

```
##           df      AIC
## buds_pois_R_04 25 12602.13
## buds_pois_R_06 26 12604.13
## buds_pois_R_07 27 12606.13
## buds_pois_R_02 27 12616.80
## buds_pois_R_03 25 12625.02
## buds_pois_R_01 25 12625.68
## buds_pois_R_05 26 12626.02
## buds_pois_R_00 24 12628.14
```

The model with the replicate id as a random effect (buds_pois_R_04) has the lowest AIC, this random effect structure is therefore selected.

3.2 Fixed effects selection

With the selected random-effect structure fixed, compare different fixed-effect structures to find the optimal one.

```
buds_pois_F_00 <- glmmTMB(buds ~ 1 + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_01 <- glmmTMB(buds ~ week_centered + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_02 <- glmmTMB(buds ~ lineage + (1 | replicate_unique), REML = FALSE, family =
poisson, data = buds_df)
buds_pois_F_03 <- glmmTMB(buds ~ sleep_condition + (1 | replicate_unique), REML = FALSE,
family = poisson, data = buds_df)
buds_pois_F_04 <- glmmTMB(buds ~ tum_state + (1 | replicate_unique), REML = FALSE, family
= poisson, data = buds_df)
buds_pois_F_05 <- glmmTMB(buds ~ week_centered + lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_06 <- glmmTMB(buds ~ week_centered + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_07 <- glmmTMB(buds ~ week_centered + tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_08 <- glmmTMB(buds ~ lineage + sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_09 <- glmmTMB(buds ~ lineage + tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_10 <- glmmTMB(buds ~ sleep_condition + tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_11 <- glmmTMB(buds ~ week_centered * lineage + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_12 <- glmmTMB(buds ~ week_centered * sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_13 <- glmmTMB(buds ~ week_centered * tum_state + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
buds_pois_F_14 <- glmmTMB(buds ~ lineage * sleep_condition + (1 | replicate_unique), REML
= FALSE, family = poisson, data = buds_df)
```

```

buds_pois_F_15 <- glmmTMB(buds ~ lineage * tum_state + (1 | replicate_unique), REML =
FALSE, family = poisson, data = buds_df)
buds_pois_F_16 <- glmmTMB(buds ~ sleep_condition * tum_state + (1 | replicate_unique),
REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_17 <- glmmTMB(buds ~ week_centered + lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_18 <- glmmTMB(buds ~ week_centered + lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_19 <- glmmTMB(buds ~ week_centered + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_20 <- glmmTMB(buds ~ lineage + sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_21 <- glmmTMB(buds ~ week_centered * lineage + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_22 <- glmmTMB(buds ~ week_centered * sleep_condition + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_23 <- glmmTMB(buds ~ week_centered * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_24 <- glmmTMB(buds ~ week_centered * lineage + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_25 <- glmmTMB(buds ~ week_centered * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_26 <- glmmTMB(buds ~ week_centered * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_27 <- glmmTMB(buds ~ lineage * sleep_condition + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_28 <- glmmTMB(buds ~ lineage * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_29 <- glmmTMB(buds ~ sleep_condition * tum_state + week_centered + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_30 <- glmmTMB(buds ~ lineage * sleep_condition + tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_31 <- glmmTMB(buds ~ lineage * tum_state + sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_32 <- glmmTMB(buds ~ sleep_condition * tum_state + lineage + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_33 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_34 <- glmmTMB(buds ~ week_centered * lineage * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_35 <- glmmTMB(buds ~ week_centered * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)
buds_pois_F_36 <- glmmTMB(buds ~ lineage * sleep_condition * tum_state + (1 |
replicate_unique), REML = FALSE, family = poisson, data = buds_df)

AIC(buds_pois_F_00, buds_pois_F_01, buds_pois_F_02, buds_pois_F_03, buds_pois_F_04,
buds_pois_F_05, buds_pois_F_06, buds_pois_F_07, buds_pois_F_08, buds_pois_F_09,
buds_pois_F_10, buds_pois_F_11, buds_pois_F_12, buds_pois_F_13, buds_pois_F_14,
buds_pois_F_15, buds_pois_F_16, buds_pois_F_17, buds_pois_F_18, buds_pois_F_19,
buds_pois_F_20, buds_pois_F_21, buds_pois_F_22, buds_pois_F_23, buds_pois_F_24,
buds_pois_F_25, buds_pois_F_26, buds_pois_F_27, buds_pois_F_28, buds_pois_F_29,
buds_pois_F_30, buds_pois_F_31, buds_pois_F_32, buds_pois_F_33, buds_pois_F_34,
buds_pois_F_35, buds_pois_F_36) %>% arrange(AIC)

```

##		df	AIC
##	buds_pois_F_33	25	12489.87
##	buds_pois_F_21	14	12494.81
##	buds_pois_F_11	13	12500.31
##	buds_pois_F_24	14	12500.54
##	buds_pois_F_34	19	12509.36
##	buds_pois_F_27	14	12561.96
##	buds_pois_F_22	10	12563.02
##	buds_pois_F_17	9	12564.93
##	buds_pois_F_05	8	12570.68
##	buds_pois_F_18	9	12570.97
##	buds_pois_F_23	10	12572.75
##	buds_pois_F_28	11	12574.36
##	buds_pois_F_14	13	12862.38
##	buds_pois_F_30	14	12862.49
##	buds_pois_F_08	8	12866.84
##	buds_pois_F_32	10	12867.85
##	buds_pois_F_20	9	12868.47
##	buds_pois_F_31	11	12870.53
##	buds_pois_F_36	19	12871.30
##	buds_pois_F_02	7	12874.06
##	buds_pois_F_09	8	12874.72
##	buds_pois_F_15	10	12878.25
##	buds_pois_F_25	6	12881.20
##	buds_pois_F_29	6	12882.98
##	buds_pois_F_19	5	12883.40
##	buds_pois_F_35	9	12883.43
##	buds_pois_F_26	6	12885.35
##	buds_pois_F_07	4	12893.10
##	buds_pois_F_12	5	12894.87
##	buds_pois_F_13	5	12895.04
##	buds_pois_F_06	4	12896.94
##	buds_pois_F_01	3	12902.29
##	buds_pois_F_16	5	13173.44
##	buds_pois_F_10	4	13174.39
##	buds_pois_F_04	3	13186.02
##	buds_pois_F_03	3	13189.04
##	buds_pois_F_00	2	13195.82

The model with the interaction between the week, the lineage and the sleep condition has the lowest AIC and is therefore selected (buds_pois_F_33).

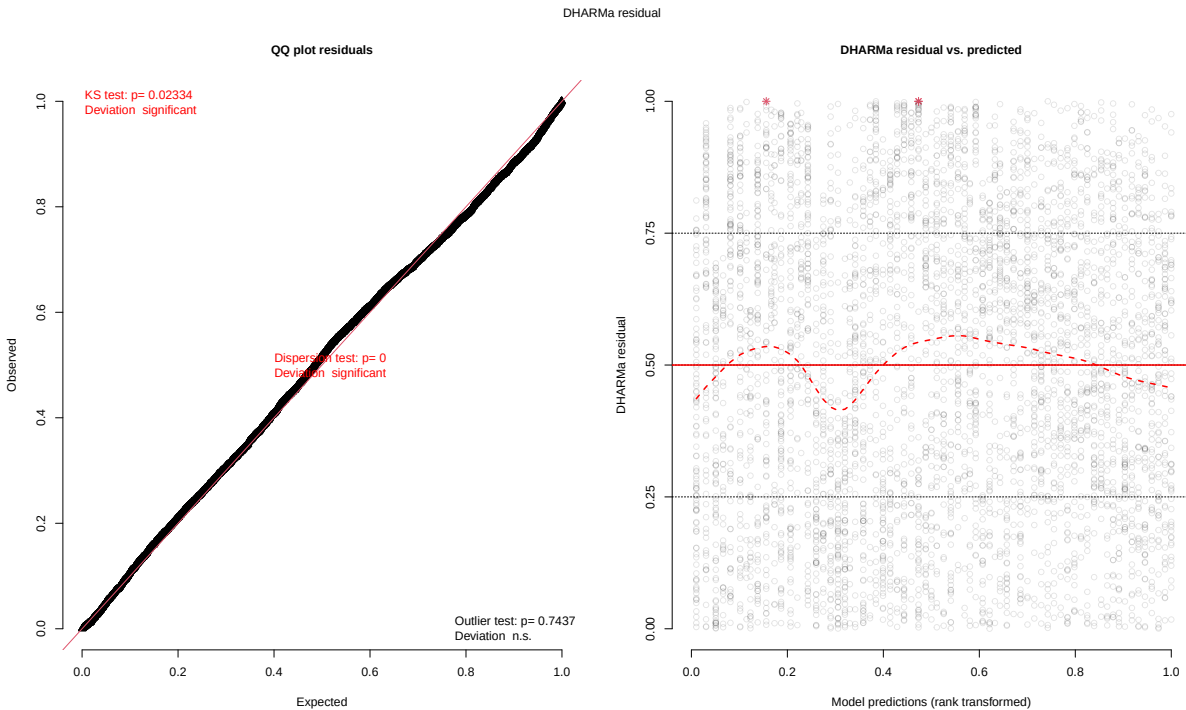
3.3 Goodness-of-fit evaluation

Refit the selected model.

```
buds_pois_mod_33 <- glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df)
```

Assess goodness-of-fit of the selected model using simulated residuals.

```
buds_pois_mod_33_res <- simulateResiduals(buds_pois_mod_33, plot = TRUE)
```



3.4 Selected model summary

Releveling of "lineage" to obtain all pairwise comparisons from the same model.

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_MT")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8   -6276.1   12552.1     4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
```

##	Estimate	
## (Intercept)	0.457269	
## week_centered	0.040455	
## lineageHC_MT	-0.573663	
## lineageHO_SPC	-0.256052	
## lineageHO_SPT	-0.092013	
## lineageHO_VLN	-1.013640	
## lineageHV_GAL	-0.061518	
## sleep_conditionSleep deprivation	-0.063490	
## week_centered:lineageHC_MT	0.227553	
## week_centered:lineageHO_SPC	0.101016	
## week_centered:lineageHO_SPT	0.126876	
## week_centered:lineageHO_VLN	0.178343	
## week_centered:lineageHV_GAL	0.140067	
## week_centered:sleep_conditionSleep deprivation	-0.007714	
## lineageHC_MT:sleep_conditionSleep deprivation	-0.209042	
## lineageHO_SPC:sleep_conditionSleep deprivation	0.140629	
## lineageHO_SPT:sleep_conditionSleep deprivation	0.030166	
## lineageHO_VLN:sleep_conditionSleep deprivation	-0.119088	
## lineageHV_GAL:sleep_conditionSleep deprivation	-0.061672	
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation	-0.007118	
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation	-0.065524	
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation	0.042134	
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation	0.034530	
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation	-0.098026	
##	Std. Error	z value
## (Intercept)	0.044793	10.209
## week_centered	0.023019	1.757
## lineageHC_MT	0.086495	-6.632
## lineageHO_SPC	0.066325	-3.861
## lineageHO_SPT	0.066574	-1.382
## lineageHO_VLN	0.074958	-13.523
## lineageHV_GAL	0.064093	-0.960
## sleep_conditionSleep deprivation	0.057329	-1.107
## week_centered:lineageHC_MT	0.044824	5.077
## week_centered:lineageHO_SPC	0.034484	2.929
## week_centered:lineageHO_SPT	0.034150	3.715
## week_centered:lineageHO_VLN	0.040251	4.431
## week_centered:lineageHV_GAL	0.033253	4.212
## week_centered:sleep_conditionSleep deprivation	0.033359	-0.231
## lineageHC_MT:sleep_conditionSleep deprivation	0.125988	-1.659
## lineageHO_SPC:sleep_conditionSleep deprivation	0.083929	1.676
## lineageHO_SPT:sleep_conditionSleep deprivation	0.084802	0.356
## lineageHO_VLN:sleep_conditionSleep deprivation	0.103689	-1.149
## lineageHV_GAL:sleep_conditionSleep deprivation	0.084267	-0.732
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation	0.069369	-0.103
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation	0.048808	-1.343
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation	0.048697	0.865
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation	0.059922	0.576
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation	0.049032	-1.999
##	Pr(> z)	
## (Intercept)	< 2e-16 ***	
## week_centered	0.078845 .	
## lineageHC_MT	3.30e-11 ***	

```
## lineageHO_SPC 0.000113 ***
## lineageHO_SPT 0.166935
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.337142
## sleep_conditionSleep deprivation 0.268097
## week_centered:lineageHC_MT 3.84e-07 ***
## week_centered:lineageHO_SPC 0.003397 **
## week_centered:lineageHO_SPT 0.000203 ***
## week_centered:lineageHO_VLN 9.39e-06 ***
## week_centered:lineageHV_GAL 2.53e-05 ***
## week_centered:sleep_conditionSleep deprivation 0.817126
## lineageHC_MT:sleep_conditionSleep deprivation 0.097072 .
## lineageHO_SPC:sleep_conditionSleep deprivation 0.093821 .
## lineageHO_SPT:sleep_conditionSleep deprivation 0.722053
## lineageHO_VLN:sleep_conditionSleep deprivation 0.250756
## lineageHV_GAL:sleep_conditionSleep deprivation 0.464253
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.918272
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.179432
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.386910
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.564451
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.045582 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPC")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) 0.20122
## week_centered 0.14147
## lineageHO_MT 0.25605
## lineageHC_MT -0.31761
## lineageHO_SPT 0.16404
## lineageHO_VLN -0.75759
## lineageHV_GAL 0.19453
## sleep_conditionSleep deprivation 0.07714
```

## week_centered:lineageHO_MT	-0.10102	
## week_centered:lineageHC_MT	0.12654	
## week_centered:lineageHO_SPT	0.02586	
## week_centered:lineageHO_VLN	0.07733	
## week_centered:lineageHV_GAL	0.03905	
## week_centered:sleep_conditionSleep deprivation	-0.07324	
## lineageHO_MT:sleep_conditionSleep deprivation	-0.14063	
## lineageHC_MT:sleep_conditionSleep deprivation	-0.34967	
## lineageHO_SPT:sleep_conditionSleep deprivation	-0.11046	
## lineageHO_VLN:sleep_conditionSleep deprivation	-0.25972	
## lineageHV_GAL:sleep_conditionSleep deprivation	-0.20230	
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation	0.06552	
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation	0.05841	
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation	0.10766	
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation	0.10005	
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation	-0.03250	
##		Std. Error z value
## (Intercept)	0.04895	4.111
## week_centered	0.02568	5.510
## lineageHO_MT	0.06633	3.861
## lineageHC_MT	0.08872	-3.580
## lineageHO_SPT	0.06944	2.362
## lineageHO_VLN	0.07751	-9.774
## lineageHV_GAL	0.06706	2.901
## sleep_conditionSleep deprivation	0.06130	1.258
## week_centered:lineageHO_MT	0.03448	-2.929
## week_centered:lineageHC_MT	0.04624	2.736
## week_centered:lineageHO_SPT	0.03599	0.718
## week_centered:lineageHO_VLN	0.04183	1.849
## week_centered:lineageHV_GAL	0.03514	1.111
## week_centered:sleep_conditionSleep deprivation	0.03563	-2.056
## lineageHO_MT:sleep_conditionSleep deprivation	0.08393	-1.676
## lineageHC_MT:sleep_conditionSleep deprivation	0.12785	-2.735
## lineageHO_SPT:sleep_conditionSleep deprivation	0.08753	-1.262
## lineageHO_VLN:sleep_conditionSleep deprivation	0.10594	-2.452
## lineageHV_GAL:sleep_conditionSleep deprivation	0.08701	-2.325
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation	0.04881	1.343
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation	0.07049	0.829
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation	0.05028	2.141
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation	0.06121	1.634
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation	0.05060	-0.642
##		Pr(> z)
## (Intercept)	3.94e-05	***
## week_centered	3.59e-08	***
## lineageHO_MT	0.000113	***
## lineageHC_MT	0.000344	***
## lineageHO_SPT	0.018155	*
## lineageHO_VLN	< 2e-16	***
## lineageHV_GAL	0.003721	**
## sleep_conditionSleep deprivation	0.208241	
## week_centered:lineageHO_MT	0.003397	**
## week_centered:lineageHC_MT	0.006213	**
## week_centered:lineageHO_SPT	0.472475	
## week_centered:lineageHO_VLN	0.064496	.

```
## week_centered:lineageHV_GAL 0.266492
## week_centered:sleep_conditionSleep deprivation 0.039815 *
## lineageHO_MT:sleep_conditionSleep deprivation 0.093821 .
## lineageHC_MT:sleep_conditionSleep deprivation 0.006238 **
## lineageHO_SPT:sleep_conditionSleep deprivation 0.206966
## lineageHO_VLN:sleep_conditionSleep deprivation 0.014220 *
## lineageHV_GAL:sleep_conditionSleep deprivation 0.020070 *
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.179432
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.407332
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.032252 *
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.102159
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.520692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_SPT")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups      Name      Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) 0.365256
## week_centered 0.167331
## lineageHO_SPC -0.164039
## lineageHO_MT 0.092013
## lineageHC_MT -0.481650
## lineageHO_VLN -0.921627
## lineageHV_GAL 0.030494
## sleep_conditionSleep deprivation -0.033324
## week_centered:lineageHO_SPC -0.025860
## week_centered:lineageHO_MT -0.126876
## week_centered:lineageHC_MT 0.100677
## week_centered:lineageHO_VLN 0.051467
## week_centered:lineageHV_GAL 0.013190
## week_centered:sleep_conditionSleep deprivation 0.034420
## lineageHO_SPC:sleep_conditionSleep deprivation 0.110464
## lineageHO_MT:sleep_conditionSleep deprivation -0.030166
## lineageHC_MT:sleep_conditionSleep deprivation -0.239207
```

```

## lineageHO_VLN:sleep_conditionSleep deprivation -0.149254
## lineageHV_GAL:sleep_conditionSleep deprivation -0.091837
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation -0.107659
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation -0.042134
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation -0.049252
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation -0.007605
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation -0.140161
##
## Std. Error z value
## (Intercept) 0.049287 7.411
## week_centered 0.025225 6.633
## lineageHO_SPC 0.069436 -2.362
## lineageHO_MT 0.066574 1.382
## lineageHC_MT 0.088903 -5.418
## lineageHO_VLN 0.077724 -11.858
## lineageHV_GAL 0.067302 0.453
## sleep_conditionSleep deprivation 0.062486 -0.533
## week_centered:lineageHO_SPC 0.035994 -0.718
## week_centered:lineageHO_MT 0.034150 -3.715
## week_centered:lineageHC_MT 0.045995 2.189
## week_centered:lineageHO_VLN 0.041552 1.239
## week_centered:lineageHV_GAL 0.034816 0.379
## week_centered:sleep_conditionSleep deprivation 0.035476 0.970
## lineageHO_SPC:sleep_conditionSleep deprivation 0.087534 1.262
## lineageHO_MT:sleep_conditionSleep deprivation 0.084802 -0.356
## lineageHC_MT:sleep_conditionSleep deprivation 0.128442 -1.862
## lineageHO_VLN:sleep_conditionSleep deprivation 0.106626 -1.400
## lineageHV_GAL:sleep_conditionSleep deprivation 0.087829 -1.046
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.050278 -2.141
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.048697 -0.865
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.070412 -0.699
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.061126 -0.124
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.050497 -2.776
##
## Pr(>|z|)
## (Intercept) 1.26e-13 ***
## week_centered 3.28e-11 ***
## lineageHO_SPC 0.018155 *
## lineageHO_MT 0.166935
## lineageHC_MT 6.04e-08 ***
## lineageHO_VLN < 2e-16 ***
## lineageHV_GAL 0.650481
## sleep_conditionSleep deprivation 0.593822
## week_centered:lineageHO_SPC 0.472475
## week_centered:lineageHO_MT 0.000203 ***
## week_centered:lineageHC_MT 0.028607 *
## week_centered:lineageHO_VLN 0.215487
## week_centered:lineageHV_GAL 0.704790
## week_centered:sleep_conditionSleep deprivation 0.331927
## lineageHO_SPC:sleep_conditionSleep deprivation 0.206966
## lineageHO_MT:sleep_conditionSleep deprivation 0.722053
## lineageHC_MT:sleep_conditionSleep deprivation 0.062549 .
## lineageHO_VLN:sleep_conditionSleep deprivation 0.161578
## lineageHV_GAL:sleep_conditionSleep deprivation 0.295727
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.032252 *
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.386910

```

```
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.484246
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.900992
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.005510 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HO_VLN")
summary(glmmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
## 12602.1 12763.8 -6276.1 12552.1    4728
##
## Random effects:
##
## Conditional model:
## Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept) -0.556370
## week_centered 0.218798
## lineageHO_SPT 0.921627
## lineageHO_SPC 0.757588
## lineageHO_MT 1.013640
## lineageHC_MT 0.439977
## lineageHV_GAL 0.952121
## sleep_conditionSleep deprivation -0.182578
## week_centered:lineageHO_SPT -0.051467
## week_centered:lineageHO_SPC -0.077327
## week_centered:lineageHO_MT -0.178343
## week_centered:lineageHC_MT 0.049210
## week_centered:lineageHV_GAL -0.038276
## week_centered:sleep_conditionSleep deprivation 0.026816
## lineageHO_SPT:sleep_conditionSleep deprivation 0.149254
## lineageHO_SPC:sleep_conditionSleep deprivation 0.259718
## lineageHO_MT:sleep_conditionSleep deprivation 0.119088
## lineageHC_MT:sleep_conditionSleep deprivation -0.089953
## lineageHV_GAL:sleep_conditionSleep deprivation 0.057417
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.007605
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation -0.100054
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation -0.034530
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation -0.041648
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation -0.132556
##
## Std. Error z value
## (Intercept) 0.060131 -9.253
```

```

## week_centered 0.033019 6.626
## lineageHO_SPT 0.077724 11.858
## lineageHO_SPC 0.077511 9.774
## lineageHO_MT 0.074958 13.523
## lineageHC_MT 0.095343 4.615
## lineageHV_GAL 0.075608 12.593
## sleep_conditionSleep deprivation 0.086399 -2.113
## week_centered:lineageHO_SPT 0.041552 -1.239
## week_centered:lineageHO_SPC 0.041827 -1.849
## week_centered:lineageHO_MT 0.040251 -4.431
## week_centered:lineageHC_MT 0.050688 0.971
## week_centered:lineageHV_GAL 0.040814 -0.938
## week_centered:sleep_conditionSleep deprivation 0.049778 0.539
## lineageHO_SPT:sleep_conditionSleep deprivation 0.106626 1.400
## lineageHO_SPC:sleep_conditionSleep deprivation 0.105936 2.452
## lineageHO_MT:sleep_conditionSleep deprivation 0.103689 1.149
## lineageHC_MT:sleep_conditionSleep deprivation 0.141613 -0.635
## lineageHV_GAL:sleep_conditionSleep deprivation 0.106192 0.541
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.061126 0.124
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.061215 -1.634
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.059922 -0.576
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.078592 -0.530
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.061383 -2.159
## Pr(>|z|)
## (Intercept) < 2e-16 ***
## week_centered 3.44e-11 ***
## lineageHO_SPT < 2e-16 ***
## lineageHO_SPC < 2e-16 ***
## lineageHO_MT < 2e-16 ***
## lineageHC_MT 3.94e-06 ***
## lineageHV_GAL < 2e-16 ***
## sleep_conditionSleep deprivation 0.0346 *
## week_centered:lineageHO_SPT 0.2155
## week_centered:lineageHO_SPC 0.0645 .
## week_centered:lineageHO_MT 9.39e-06 ***
## week_centered:lineageHC_MT 0.3316
## week_centered:lineageHV_GAL 0.3483
## week_centered:sleep_conditionSleep deprivation 0.5901
## lineageHO_SPT:sleep_conditionSleep deprivation 0.1616
## lineageHO_SPC:sleep_conditionSleep deprivation 0.0142 *
## lineageHO_MT:sleep_conditionSleep deprivation 0.2508
## lineageHC_MT:sleep_conditionSleep deprivation 0.5253
## lineageHV_GAL:sleep_conditionSleep deprivation 0.5887
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.9010
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.1022
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.5645
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.5962
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.0308 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

buds_df$lineage <- relevel(buds_df$lineage, ref = "HC_MT")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))

```

```

## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##           AIC           BIC      logLik -2*log(L)  df.resid
##    12602.1    12763.8    -6276.1    12552.1      4728
##
## Random effects:
##
## Conditional model:
## Groups           Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111  0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
##                                     Estimate
## (Intercept)                        -0.116393
## week_centered                      0.268008
## lineageHO_VLN                     -0.439977
## lineageHO_SPT                      0.481650
## lineageHO_SPC                      0.317611
## lineageHO_MT                      0.573663
## lineageHV_GAL                     0.512144
## sleep_conditionSleep deprivation -0.272531
## week_centered:lineageHO_VLN        -0.049210
## week_centered:lineageHO_SPT        -0.100677
## week_centered:lineageHO_SPC        -0.126537
## week_centered:lineageHO_MT         -0.227553
## week_centered:lineageHV_GAL        -0.087487
## week_centered:sleep_conditionSleep deprivation -0.014832
## lineageHO_VLN:sleep_conditionSleep deprivation 0.089953
## lineageHO_SPT:sleep_conditionSleep deprivation 0.239207
## lineageHO_SPC:sleep_conditionSleep deprivation 0.349671
## lineageHO_MT:sleep_conditionSleep deprivation 0.209042
## lineageHV_GAL:sleep_conditionSleep deprivation 0.147370
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.041648
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.049252
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation -0.058406
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.007118
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation -0.090908
##                                     Std. Error z value
## (Intercept)                        0.074015 -1.573
## week_centered                      0.038461  6.968
## lineageHO_VLN                     0.095343 -4.615
## lineageHO_SPT                      0.088903  5.418
## lineageHO_SPC                      0.088717  3.580
## lineageHO_MT                      0.086495  6.632
## lineageHV_GAL                     0.087060  5.883
## sleep_conditionSleep deprivation 0.112202 -2.429
## week_centered:lineageHO_VLN        0.050688 -0.971
## week_centered:lineageHO_SPT        0.045995 -2.189

```

```
## week_centered:lineageHO_SPC 0.046244 -2.736
## week_centered:lineageHO_MT 0.044824 -5.077
## week_centered:lineageHV_GAL 0.045329 -1.930
## week_centered:sleep_conditionSleep deprivation 0.060821 -0.244
## lineageHO_VLN:sleep_conditionSleep deprivation 0.141613 0.635
## lineageHO_SPT:sleep_conditionSleep deprivation 0.128442 1.862
## lineageHO_SPC:sleep_conditionSleep deprivation 0.127850 2.735
## lineageHO_MT:sleep_conditionSleep deprivation 0.125988 1.659
## lineageHV_GAL:sleep_conditionSleep deprivation 0.128148 1.150
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.078592 0.530
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.070412 0.699
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.070488 -0.829
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.069369 0.103
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.070639 -1.287
## Pr(>|z|)
## (Intercept) 0.115820
## week_centered 3.21e-12 ***
## lineageHO_VLN 3.94e-06 ***
## lineageHO_SPT 6.04e-08 ***
## lineageHO_SPC 0.000344 ***
## lineageHO_MT 3.30e-11 ***
## lineageHV_GAL 4.04e-09 ***
## sleep_conditionSleep deprivation 0.015143 *
## week_centered:lineageHO_VLN 0.331626
## week_centered:lineageHO_SPT 0.028607 *
## week_centered:lineageHO_SPC 0.006213 **
## week_centered:lineageHO_MT 3.84e-07 ***
## week_centered:lineageHV_GAL 0.053602 .
## week_centered:sleep_conditionSleep deprivation 0.807336
## lineageHO_VLN:sleep_conditionSleep deprivation 0.525295
## lineageHO_SPT:sleep_conditionSleep deprivation 0.062549 .
## lineageHO_SPC:sleep_conditionSleep deprivation 0.006238 **
## lineageHO_MT:sleep_conditionSleep deprivation 0.097072 .
## lineageHV_GAL:sleep_conditionSleep deprivation 0.250145
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.596166
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.484246
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.407332
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.918272
## week_centered:lineageHV_GAL:sleep_conditionSleep deprivation 0.198115
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
buds_df$lineage <- relevel(buds_df$lineage, ref = "HV_GAL")
summary(glmTMB(buds ~ week_centered * lineage * sleep_condition + (1 |
replicate_unique), REML = TRUE, family = poisson, data = buds_df))
```

```
## Family: poisson ( log )
## Formula:
## buds ~ week_centered * lineage * sleep_condition + (1 | replicate_unique)
## Data: buds_df
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##  12602.1  12763.8  -6276.1   12552.1     4728
```

```

##
## Random effects:
##
## Conditional model:
##   Groups          Name          Variance Std.Dev.
## replicate_unique (Intercept) 0.03111 0.1764
## Number of obs: 4753, groups: replicate_unique, 428
##
## Conditional model:
##
## (Intercept)                                Estimate
## week_centered                                0.18052
## lineageHC_MT                                -0.51214
## lineageHO_VLN                               -0.95212
## lineageHO_SPT                               -0.03049
## lineageHO_SPC                               -0.19453
## lineageHO_MT                                 0.06152
## sleep_conditionSleep deprivation            -0.12516
## week_centered:lineageHC_MT                   0.08749
## week_centered:lineageHO_VLN                  0.03828
## week_centered:lineageHO_SPT                 -0.01319
## week_centered:lineageHO_SPC                 -0.03905
## week_centered:lineageHO_MT                 -0.14007
## week_centered:sleep_conditionSleep deprivation -0.10574
## lineageHC_MT:sleep_conditionSleep deprivation -0.14737
## lineageHO_VLN:sleep_conditionSleep deprivation -0.05742
## lineageHO_SPT:sleep_conditionSleep deprivation 0.09184
## lineageHO_SPC:sleep_conditionSleep deprivation 0.20230
## lineageHO_MT:sleep_conditionSleep deprivation 0.06167
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation 0.09091
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation 0.13256
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation 0.14016
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation 0.03250
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation 0.09803
##
## (Intercept)                                Std. Error z value
## week_centered                                0.04592 8.618
## lineageHC_MT                                0.08706 -5.883
## lineageHO_VLN                               0.07561 -12.593
## lineageHO_SPT                               0.06730 -0.453
## lineageHO_SPC                               0.06706 -2.901
## lineageHO_MT                                0.06409 0.960
## sleep_conditionSleep deprivation            0.06174 -2.027
## week_centered:lineageHC_MT                   0.04533 1.930
## week_centered:lineageHO_VLN                  0.04081 0.938
## week_centered:lineageHO_SPT                 0.03482 -0.379
## week_centered:lineageHO_SPC                 0.03514 -1.111
## week_centered:lineageHO_MT                 0.03325 -4.212
## week_centered:sleep_conditionSleep deprivation 0.03593 -2.943
## lineageHC_MT:sleep_conditionSleep deprivation 0.12815 -1.150
## lineageHO_VLN:sleep_conditionSleep deprivation 0.10619 -0.541
## lineageHO_SPT:sleep_conditionSleep deprivation 0.08783 1.046
## lineageHO_SPC:sleep_conditionSleep deprivation 0.08701 2.325
## lineageHO_MT:sleep_conditionSleep deprivation 0.08427 0.732

```

```
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation    0.07064    1.287
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation    0.06138    2.159
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation    0.05050    2.776
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation    0.05060    0.642
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation    0.04903    1.999
##                                                                    Pr(>|z|)
## (Intercept)                                                       < 2e-16 ***
## week_centered                                                       5.38e-14 ***
## lineageHC_MT                                                       4.04e-09 ***
## lineageHO_VLN                                                       < 2e-16 ***
## lineageHO_SPT                                                       0.65048
## lineageHO_SPC                                                       0.00372 **
## lineageHO_MT                                                       0.33714
## sleep_conditionSleep deprivation                                  0.04265 *
## week_centered:lineageHC_MT                                          0.05360 .
## week_centered:lineageHO_VLN                                         0.34833
## week_centered:lineageHO_SPT                                         0.70479
## week_centered:lineageHO_SPC                                         0.26649
## week_centered:lineageHO_MT                                          2.53e-05 ***
## week_centered:sleep_conditionSleep deprivation                    0.00325 **
## lineageHC_MT:sleep_conditionSleep deprivation                     0.25015
## lineageHO_VLN:sleep_conditionSleep deprivation                    0.58872
## lineageHO_SPT:sleep_conditionSleep deprivation                     0.29573
## lineageHO_SPC:sleep_conditionSleep deprivation                     0.02007 *
## lineageHO_MT:sleep_conditionSleep deprivation                      0.46425
## week_centered:lineageHC_MT:sleep_conditionSleep deprivation        0.19812
## week_centered:lineageHO_VLN:sleep_conditionSleep deprivation        0.03081 *
## week_centered:lineageHO_SPT:sleep_conditionSleep deprivation        0.00551 **
## week_centered:lineageHO_SPC:sleep_conditionSleep deprivation        0.52069
## week_centered:lineageHO_MT:sleep_conditionSleep deprivation        0.04558 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4 Estimated marginal means

4.1 Table

Create a character string containing the 6 exact centered values.

```
week_values <- 1:6 - mean(buds_df$week, na.rm = TRUE)
week_terms <- paste0("week_centered [", paste(week_values, collapse = ","), "]")
```

Compute the estimated marginal means of weekly bud production for every combination of week, sleep condition, and lineage, based on the selected model.

```
buds_emmeans <-
  as.data.frame(ggemmeans(buds_pois_mod_33,
    terms = c(week_terms, "sleep_condition", "lineage"),
    bias_correction = TRUE)) %>%
  arrange(facet, group, x)
```

Back-transform the centered week to the original scale (weeks 1 to 6) for plotting.

```
buds_emmeans$x <- buds_emmeans$x + mean(buds_df$week, na.rm = TRUE)
```

Round the values to ensure ggplot perfectly recognizes the integers 1 to 6.

```
buds_emmeans$x <- round(buds_emmeans$x)
```

4.2 Plots

4.3 Common functions

Function to plot, for a single lineage, the predicted weekly bud production by sleep condition.

```
buds_plot_treatment <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)

  ggplot(data_sub, aes(x = x, y = predicted, color = group, fill = group)) +
    geom_line(linewidth = 0.8) +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
    labs(x = if(show_x_title) "Age (weeks)" else NULL,
         y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
         else NULL,
         title = NULL) +
    theme_bw() +
    theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
          element_blank(),
          axis.ticks.x = if(show_x_text) element_line() else element_blank(),
          axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
          element_blank(),
          axis.ticks.y = if(show_y_text) element_line() else element_blank(),
          axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
            10)),
          axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
            margin = margin(l = 25)),
          axis.line.y.right = element_blank(),
          plot.margin = margin(l = 5, r = 5, t = 5, b = 5),
          legend.position = "none") +
    scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
    scale_y_continuous(position = "right") +
    coord_cartesian(ylim = c(0, 3.5)) +
    scale_color_manual(values = sleep_condition_colors) +
    scale_fill_manual(values = sleep_condition_colors) +
    annotate("text", x = 1.2, y = 3.3, label = lineage,
            color = "black", fontface = "bold", size = 5, hjust = 0, vjust = 1)}
```

Function to plot, for a single lineage, the predicted weekly bud production pooled across sleep conditions.

```
buds_plot_lineage <- function(lineage, show_y_text = FALSE, show_x_text = TRUE,
show_y_title = FALSE, show_x_title = FALSE) {
  data_sub <- subset(buds_emmeans, facet == lineage)
  data_lineage <- aggregate(cbind(predicted, conf.low, conf.high) ~ x, data = data_sub,
FUN = mean)
```

```

data_lineage <- data_lineage[order(data_lineage$x), ]

ggplot(data_lineage, aes(x = x, y = predicted)) +
  geom_line(linewidth = 0.8, color = "black") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, fill = "black",
    linewidth = 0.3) +
  labs(x = if(show_x_title) "Age (weeks)" else NULL,
    y = if(show_y_title) "Estimated marginal means\nof the number of buds produced"
    else NULL,
    title = NULL) +
  theme_bw() +
  theme(axis.text.x = if(show_x_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.x = if(show_x_text) element_line() else element_blank(),
    axis.text.y = if(show_y_text) element_text(color = "black", size = 14) else
    element_blank(),
    axis.ticks.y = if(show_y_text) element_line() else element_blank(),
    axis.title.x = element_text(color = "black", size = 14, margin = margin(t =
    10)),
    axis.title.y.right = element_text(color = "black", size = 14, angle = 90,
    margin = margin(l = 25)),
    axis.line.y.right = element_blank(),
    plot.margin = margin(l = 5, r = 5, t = 5, b = 5)) +
  scale_x_continuous(limits = c(1, 6), breaks = c(2, 4, 6)) +
  scale_y_continuous(position = "right") +
  coord_cartesian(ylim = c(0, 3.5)) +
  annotate("text", x = 1.2, y = 3.3, label = lineage,
    color = "black", fontface = "bold", size = 5, hjust = 0, vjust = 1) }

```

4.4 Panel A: interaction between sleep condition and lineage

```

A1 <- buds_plot_treatment("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
A2 <- buds_plot_treatment("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
A3 <- buds_plot_treatment("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
A4 <- buds_plot_treatment("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
A5 <- buds_plot_treatment("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title
= TRUE)
A6 <- buds_plot_treatment("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title
= TRUE)

```

Add a letter tag (A) directly to the first sub-plot before combining.

```

A1 <- A1 + labs(tag = "A") + theme(plot.tag = element_text(size = 20, face = "bold"))

```

Combine A1 to A6 into panel A.

```

A <- (A1 + A2 + A3) / (A4 + A5 + A6) + plot_layout(heights = c(1, 1))

```

4.5 Panel B: lineage effect

```
B1 <- buds_plot_lineage("HO_MT", show_y_text = FALSE, show_x_text = FALSE)
B2 <- buds_plot_lineage("HO_SPC", show_y_text = FALSE, show_x_text = FALSE)
B3 <- buds_plot_lineage("HO_SPT", show_y_text = TRUE, show_x_text = FALSE)
B4 <- buds_plot_lineage("HO_VLN", show_y_text = FALSE, show_x_text = TRUE)
B5 <- buds_plot_lineage("HC_MT", show_y_text = FALSE, show_x_text = TRUE, show_x_title = TRUE)
B6 <- buds_plot_lineage("HV_GAL", show_y_text = TRUE, show_x_text = TRUE, show_y_title = TRUE)
```

Add a letter tag (B) directly to the first sub-plot before combining.

```
B1 <- B1 + labs(tag = "B") + theme(plot.tag = element_text(size = 20, face = "bold"))
```

Combine B1 to B6 into panel B.

```
B <- (B1 + B2 + B3) / (B4 + B5 + B6) + plot_layout(heights = c(1, 1))
```

4.6 Assembly

Extract legend from the plot A6 to use later on the full figure.

```
legend_treatment <- cowplot::get_legend(
  A6 +
  labs(color = NULL, fill = NULL) +
  theme(legend.position = "bottom",
        legend.direction = "horizontal",
        legend.text = element_text(size = 13)))
```

Build title graphical objects for each panel.

```
interaction_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Interaction between sleep condition and lineage",
    fontface = "bold", size = 16, x = 0.5, hjust = 0.5)

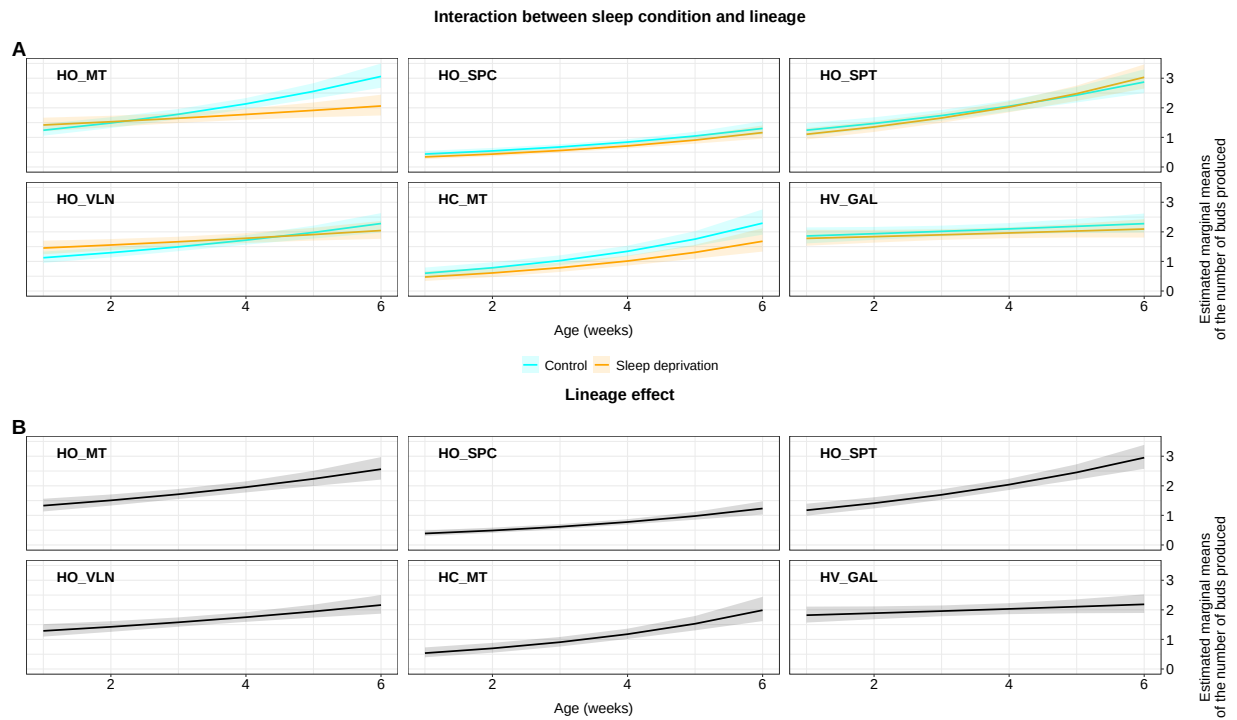
lineage_title_grob <- cowplot::ggdraw() +
  cowplot::draw_label("Lineage effect",
    fontface = "bold", size = 16, x = 0.5, hjust = 0.5)
```

Stack the titles, panels, and legend vertically into a single column.

```
final_buds_panel <- cowplot::plot_grid(interaction_title_grob,
  A,
  legend_treatment,
  lineage_title_grob,
  B,
  ncol = 1,
  rel_heights = c(0.08, 1, 0.1, 0.08, 1)) + #
  Valeurs modifiées ici (0.08)
  theme(plot.margin = margin(l = 5, r = 5, t = 5, b = 5))
```

Show the final figure.

```
print(final_buds_panel)
```



```
ggsave("figure_4.png", plot = final_buds_panel, width = 21, height = 29.7, units = "cm",
dpi = 300)
```